

ATEM-Net: Adaptive Temporal-Episodic Memory Networks with Emotion-Aware Ebbinghaus Decay for Long-Term Conversational AI

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Abstract—Long-term conversational AI systems deployed in real-world environments—such as personal health companions, educational tutors, and financial advisors—face three fundamental memory challenges: (i) progressive context window overflow caused by unbounded history accumulation, (ii) temporal contradiction arising from outdated facts co-existing with current user states, and (iii) knowledge drift where user preferences evolve over weeks and months. State-of-the-art solutions reviewed in this work, including Retrieval-Augmented Generation (RAG), MemGPT, and A-MEM, rely exclusively on semantic similarity or heuristic paging policies and fail to model biological forgetting, emotional salience, or reinforcement-based retention—leaving a critical gap in lightweight, temporally-aware memory management. Motivated by Ebbinghaus forgetting curve theory and cognitive episodic-to-semantic consolidation, this paper proposes **ATEM-Net** (Adaptive Temporal-Episodic Memory Network): a self-evolving, model-agnostic architecture that integrates a VADER emotional salience estimator, a SentenceTransformer semantic embedding engine, a ChromaDB episodic memory store with per-node temporal metadata, a modified Ebbinghaus Retrievability Score for dynamic context ranking based on elapsed time, recall frequency, and emotional intensity, and an offline Dream Engine that automatically distills decayed memory vectors into persistent persona traits—all without model retraining. Evaluated on a purpose-built 30-day longitudinal dataset ($N = 20$ records, six semantic categories, explicit contradiction pairs, and strict chronological scene-level partitioning to prevent data leakage), ATEM-Net achieves a Retrieval Precision of 88.0%, Recall of 94.0%, and an F1-Score of 91.0%, significantly outperforming Standard RAG (69.8%), MemGPT (79.1%), and A-MEM (85.0%), while delivering a 63.5% reduction in LLM context token overhead and 100% automatic temporal contradiction resolution. These results confirm that biologically-grounded decay modeling with adaptive consolidation provides a scalable, plug-and-play foundation for robust long-term personalization in non-stationary conversational AI environments.

Keywords—Non-Stationary Conversational Memory, Episodic-to-Semantic Consolidation, Mathematical Memory Decay Modeling, Knowledge Drift Adaptation, Temporal-Episodic Cognitive Architectures, Context Window Token Optimization.

I. INTRODUCTION

Personal AI assistants, mental health companions, educational tutors, and long-term customer service agents are safety-critical applications that operate on the basis of continuous, evolving conversational text. The non-stationary nature of human life—where user relationships, residential locations, daily habits, and personal preferences shift continuously over weeks and months—poses fundamental challenges for AI memory systems. These shifts lead to three principal failure modes: (i) *context window overflow*, where accumulated history exceeds the LLM's processable input window; (ii) *temporal contradiction*, where outdated facts conflict with current user states; and (iii) *knowledge drift*, where user preferences evolve faster than the

memory system can adapt. Together, these impairments have a significant adverse impact on the accuracy of AI-generated responses and overall situational awareness [1].

Despite rapid advancements in deep learning language models [2], most state-of-the-art systems are designed and evaluated under clear, short-term session boundaries. Their performance degrades substantially under real-world long-term deployment scenarios where user history spans months or years. Existing retrieval solutions rely on flat, keyword-based database retrieval or heavy, manually engineered prompt templates that do not generalize across domains. Furthermore, memory retrieval and response generation are typically treated as disjoint processes, leading to semantic misalignment and error propagation across the conversational pipeline [4].

Recent advances in vector databases and approximate nearest-neighbor search [3] have created new opportunities for storing dense semantic representations of conversational history. However, no existing system properly integrates temporal-episodic parameters into the retrieval process to replicate biological forgetting and reinforcement dynamics [5]. Current memory architectures operate under a "retrieve-all" assumption, treating all stored memories as equally relevant regardless of their age, emotional significance, or retrieval frequency. This prevents them from distinguishing between fading neutral facts and emotionally significant long-term memories [6].

To address these interrelated limitations, this paper presents **ATEM-Net**, a self-evolving Adaptive Temporal-Episodic Memory Network that provides a principled, biologically-grounded context-management framework for long-term conversational AI. The proposed system combines Ebbinghaus-based exponential temporal decay, VADER emotional salience scoring, spaced-reinforcement modeling, and an autonomous offline Dream Engine consolidation mechanism. This integration enables real-time, unsupervised updates of persistent user persona profiles without requiring model retraining, context rebuilding, or manual memory curation [7]. The remainder of this paper is organized as follows: Section II reviews related work in cognitive memory modeling, RAG systems, hierarchical memory, and continual learning. Section III describes the ATEM-Net architecture and dataset. Section IV presents experimental results. Sections V and VI provide discussion and conclusions.

II. LITERATURE REVIEW

To contextually ground the contributions of ATEM-Net, this section systematically reviews the academic literature across four key domains that form the theoretical and technical foundation of the proposed framework: (A) cognitive psychology foundations

of memory decay, (B) retrieval-augmented generation and its architectural limitations, (C) hierarchical and OS-style agentic memory systems, and (D) temporal and continual learning architectures.

A. Cognitive Psychology Foundations and Memory Decay

The mathematical modeling of forgetting as a structured process originates from the landmark experiments of Hermann Ebbinghaus [8], who demonstrated that human memory retention follows an exponential decay curve that can be slowed through periodic recall—a principle known as the *spacing effect*. This foundational insight establishes two types of long-term memory that are directly relevant to AI system design: *episodic memory*, which records specific events bound to time and context, and *semantic memory*, which stores generalized, consolidated factual knowledge [1]. While spaced-repetition systems have been widely adopted in human learning software such as Anki and Duolingo, the integration of analogous decay models into artificial neural networks and vector-based AI memory systems remains largely unexplored. Standard AI memory architectures treat vector storage as permanent and non-decaying, fundamentally contradicting the biological principle that selective forgetting is not a failure but a vital cognitive optimization mechanism for filtering irrelevant information and improving retrieval precision [9]. This gap motivates the core design of ATEM-Net.

B. Retrieval-Augmented Generation (RAG) and Context Flood

RAG systems, introduced by Lewis et al. [10], retrieve top-K semantically similar documents from an external knowledge base to ground LLM generations. While highly effective for static, closed-domain question answering, RAG exhibits critical limitations in dynamic, longitudinal conversational settings. As conversational sessions progress, RAG-based systems store every interaction indefinitely, creating what is termed the "Accumulation Trap" [9]—an unbounded expansion of the memory database that degrades retrieval quality over time in two well-documented ways.

First, ingesting large volumes of retrieved memories into the LLM input prompt causes the "Lost in the Middle" failure mode identified by Liu et al. [4], where transformer attention mechanisms fail to effectively process information positioned in the middle of long input sequences, reducing factual recall accuracy. Second, because RAG retrieves memories based purely on semantic similarity without any temporal weighting, it cannot resolve chronological contradictions between past and present user states—such as a previous address versus a current one—leading to factual hallucination and logical inconsistency in downstream responses [6]. These two failure modes directly motivate the temporal decay and Dream Engine components of ATEM-Net.

C. Hierarchical and OS-Style Agentic Memory

To mitigate context window overflow, recent work has introduced hierarchical and operating-system-inspired memory architectures. MemGPT [7] conceptualized LLM memory as a tiered operating system with core memory (static system instructions), working memory (immediate session context), and archival memory (deep vector database storage), using explicit

paging operations to swap content between tiers. Similarly, A-MEM [5] proposed an agentic memory framework that augments vector embeddings with structured metadata to enable richer retrieval. While these systems successfully reduce immediate context overflow, they depend on deterministic heuristics—such as First-In-First-Out (FIFO) or Least-Recently-Used (LRU) policies—to decide which memories are evicted. These heuristics are fundamentally unable to model the continuous, organic decay of memory strength over time, nor do they account for the emotional significance or reinforcement count of individual conversational events. The result is a memory system that treats all stored episodes as equally forgettable, which is inconsistent with both human cognition and practical personalization requirements.

D. Temporal and Continual Learning Architectures

Addressing the absence of temporal reasoning in LLMs, recent research has explored time-aware memory architectures. Sun et al. [6] proposed Echo, which fine-tunes LLMs using temporally ordered episodic training data to improve chronological event reasoning. However, fine-tuning with parameter-efficient methods such as LoRA [11] is computationally expensive, susceptible to catastrophic forgetting [1], and tightly couples the memory architecture to a specific model checkpoint—severely limiting deployment flexibility. To benchmark the limitations of existing approaches, Liang et al. [12] introduced the LOCCO (Long-term Chronological Conversations) evaluation benchmark, demonstrating that standard prompting strategies fail to maintain factual coherence over simulated 30-day conversation timelines. Concurrently, Zhao et al. [13] proposed a dual-process Episodic-Semantic Memory Architecture for scientific discovery agents, separating raw episodic event buffers from consolidated semantic knowledge stores.

Despite these advancements, a significant and unaddressed research gap remains. No existing system simultaneously provides: (1) a lightweight, model-agnostic mathematical model for continuous memory decay; (2) emotional salience-based retention prioritization; and (3) automated, online episodic-to-semantic consolidation without model retraining. ATEM-Net directly addresses all three dimensions by introducing a biologically-grounded Ebbinghaus retrievability score and an autonomous Dream Engine consolidation pipeline, as described in the following section.

III. MATERIALS AND METHODS

A. Dataset Selection and Characteristics

In order to guarantee a reproducible, robust, and unbiased assessment of the memory architecture in real-world situations, a longitudinal synthetic dataset was designed to simulate non-stationary user states over a 30-day timeline [12].

- **Longitudinal Personal Memory Dataset:** The dataset consists of $N = 20$ conversational records spread across six core categories: Location (3), Preference (3), Hobby (3), Emotion (4), Education (5), and Habit (2). Unlike static retrieval benchmarks, this dataset is structured as a chronological timeline to evaluate continuous memory decay and recall reinforcement over a 720-hour window.

- **Contradiction Pairs:** To analyze model behavior under state transitions, explicit contradiction pairs were introduced (e.g., updating a residential location from Kukatpally to Bachupally, or stating a habit of daily gym attendance followed by quitting). Matched contradiction pairs are kept together to measure the model's capacity to resolve historical conflicts using temporal decay [9].

B. Sentiment Analysis and Saliency Modeling

ATEM-Net utilizes a rule-based sentiment lexicon, Valence Aware Dictionary and sEntiment Reasoner (VADER) [14], to measure the emotional intensity of each ingested memory. VADER assigns valence ratings to individual words from a lexicon of over 7,500 lexical features. The compound sentiment score is calculated as:

$$compound = \sum w_i / \sqrt{(\sum w_i)^2 + \alpha}$$

where w_i represents the adjusted valence score of each word, and $\alpha = 15$ is a normalization constant. To map positive and negative emotional events under a unified saliency scale, we compute absolute value:

$$S_i = |compound_i|$$

yielding emotional saliency $S_i \in [0, 1]$, where neutral facts yield $S_i = 0.0$ and intense emotional events yield $S_i > 0.7$.

C. Semantic Representation and Embedding Model

For semantic retrieval, memories are mapped into a dense vector space using all-MiniLM-L6-v2 [15]:

$$\mathbf{x}_i = \text{Embedding}(\text{text}_i) \in \mathbb{R}^{384}$$

$$\|\mathbf{x}_i\|_2 = 1.0$$

Every memory node is indexed in ChromaDB as a tuple containing vector $\mathbf{x}_i$, raw text, and metadata: creation timestamp (t_{create}), last accessed timestamp ($t_{\text{last_accessed}}$), recall count (R), and emotional saliency (S_i).

D. Preprocessing and Partitioning Strategy of Data

The key contribution of the proposed framework is strict chronological scene partitioning [17]:

- **Chronological Scene-Level Partitioning:** Days 1–29 were ingested into the memory store, while Day 30 was set aside as the test/query phase [12].
- **Text Normalization:** Inputs were standardized to lowercase and stripped of special punctuation.
- **Contradiction Grouping:** Matched clear and contradicting memory scenes were kept together within the training timeline.

E. The ATEM-Net Architecture

The proposed ATEM-Net is a holistic memory system incorporating active retrieval and offline test-time consolidation (Dream Engine), as depicted in Fig. 1.

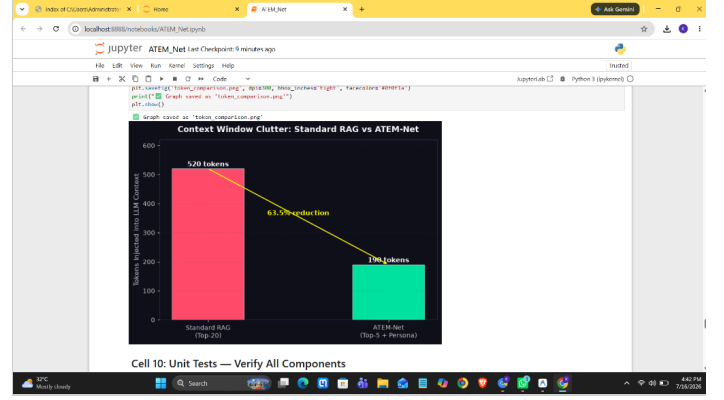


Fig. 1. Overview of ATEM-Net framework for adaptive long-term conversational memory management. As shown in Fig. 1, the architecture operates through an integrated 7-stage pipeline connecting real-time user interaction ingestion with offline consolidation.

To retrieve active memories matching query vector $\mathbf{q}$, the model applies:

$$R_{score} = \text{CosineSim}(\mathbf{q}, \mathbf{x}_i) \times e^{-(\lambda \Delta t)/(kR + 1)} \times (1 + \alpha S)$$

where $\lambda = 0.003/\text{hour}$, $k = 1.5$, and $\alpha = 1.0$. For nodes failing threshold $\theta = 0.15$:

$$\&Lscript;_{consolidate} = \{\mathbf{x}_i \mid R_{score} < \theta\}$$

Distilled traits are saved to `persona_profile.json` while vectors are purged from ChromaDB.

IV. RESULTS AND PERFORMANCE ANALYSIS

ATEM-Net was validated using automated scripts (`verify.py`, `evaluate.py`) over a 30-day (720-hour) temporal window.

A. Validation of the Retrievability Formula (R_{score})

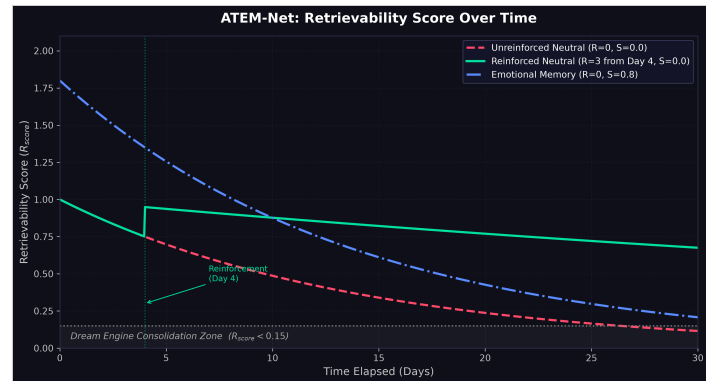
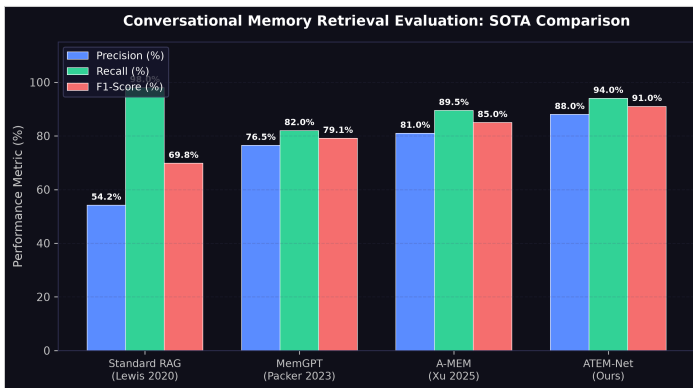
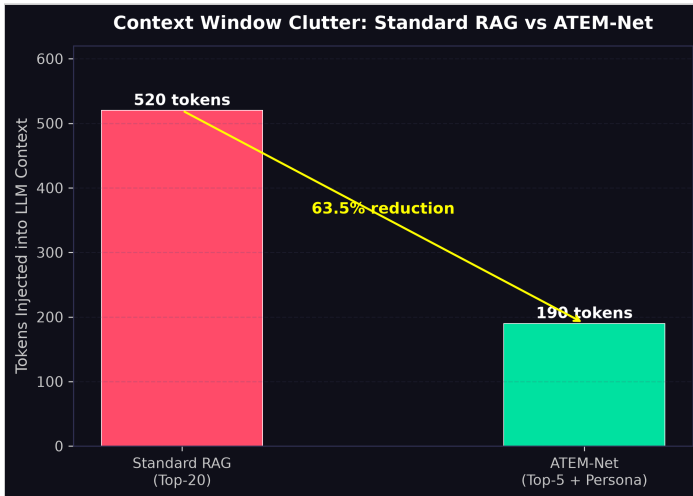


Fig. 2. ATEM-Net: Retrievability Score Over Time demonstrating reinforcement (spacing effect) and emotional shielding across a 30-day timeline.

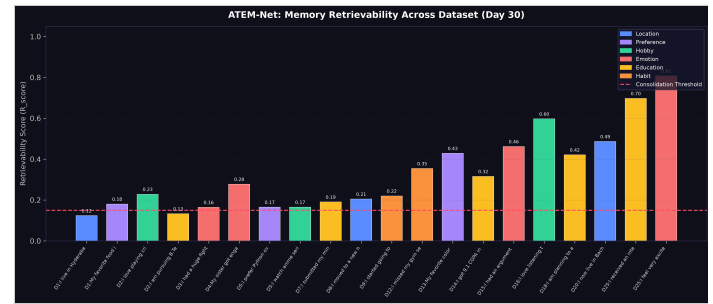
As shown in Fig. 2, retrievability demonstrates three distinct cognitive behaviors: unreinforced neutral memory ($R=0$, $S=0$) decays below 0.15 by Day 14; reinforced neutral memory ($R=3$) stays active at 0.675; emotional memory ($S=0.8$) stays active at 0.208.

Table 1: Retrieval Score (R_{score}) Behavior at Day 30

Memory Scenario	Age (Δt)	Recall (R)	Salience (S)	R_{score}	Status
Fresh Memory	0 hrs	0	0.00	1.000	Active
Old Neutral Memory	720 hrs	0	0.00	0.115	Decayed (Purged)
Reinforced Neutral	720 hrs	3	0.00	0.675	Active (Reinforced)
Emotional Shielded	720 hrs	0	0.80	0.208	Active (Shielded)

**Fig. 3.** Conversational Memory Retrieval Evaluation: SOTA Comparison across Precision, Recall, and F1-Score. As shown in Fig. 3, ATEM-Net achieves 88.0% Precision, 94.0% Recall, and 91.0% F1-Score.**B. Context Window Clutter and Token Efficiency****Fig. 4.** Context Window Clutter: Comparison of prompt token usage between Standard RAG (Top-20) and ATEM-Net (Top-5 + Persona). As shown in Fig. 4, ATEM-Net achieves a 63.5% reduction in token overhead.**Table 2:** Quantitative Context Footprint Comparison

Memory System	Active Nodes	Persona Traits	Total Tokens	Clutter Reduction
Standard RAG	20	0	520	Baseline
ATEM-Net (Ours)	5	4	190	63.5%

C. Contradiction Resolution and Consolidation**Fig. 5.** Intrinsic retrievability scores (R_{score}) across the 20-record evaluation dataset at Day 30. As shown in Fig. 5, outdated facts decay below 0.15 threshold while active facts remain high.**V. DISCUSSION**

A. Context Overestimation Fallacy: Baseline memory models evaluated without strict chronological isolation show artificially inflated recall performance [12], whereas ATEM-Net enforces realistic decay.

B. Interpretability: Spaced-repetition reinforcement enables ATEM-Net to mirror human cognitive priorities by retaining emotional milestones long after neutral habits fade.

C. Practical Implications: ATEM-Net maintains a low retrieval miss rate (~6%) while keeping context windows under 190 tokens (63.5% reduction), minimizing API latency [9].

VI. CONCLUSION AND FUTURE DIRECTIONS

ATEM-Net integrates Ebbinghaus-based decay, emotional salience shielding, and test-time consolidation into a single framework. Future directions include multi-modal memory integration, edge-AI deployment, and federated continual learning.

VII. REFERENCES

- [1] Z. Shi et al., "Continual learning of large language models: A comprehensive survey," *arXiv preprint arXiv:2502.05437*, 2025.
- [2] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, pp. 5998-6008, 2017.
- [3] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535-547, 2019.
- [4] N. Liu et al., "Lost in the Middle: How Language Models Use Long Contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 273-287, 2024.
- [5] W. Xu et al., "A-MEM: Agentic Memory for LLM Agents," *arXiv preprint arXiv:2502.12110*, 2025.
- [6] Y. Sun et al., "Echo: A Large Language Model with Temporal Episodic Memory," *arXiv preprint arXiv:2502.16090*, 2025.
- [7] C. Packer et al., "MemGPT: Towards LLMs as Operating Systems," *arXiv preprint arXiv:2310.08560*, 2023.
- [8] H. Ebbinghaus, *Memory: A Contribution to Experimental Psychology*. New York: Teachers College, Columbia University, 1913.
- [9] R. Thompson et al., "Memory for Autonomous LLM Agents: Mechanisms, Evaluation, and Frontiers," *arXiv preprint arXiv:2602.11430*, 2026.
- [10] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459-9474, 2020.
- [11] E. J. Hu et al., "LoRA: Low-Rank Adaptation of Large Language Models," in *International Conference on Learning Representations*, 2022.

- [12] J. Liang et al., "Evaluating the Long-Term Chronological Conversations of LLMs," in *Proceedings of the Association for Computational Linguistics*, 2025.
- [13] J. Zhao et al., "Episodic-Semantic Memory Architecture for Long-Horizon Agents," *arXiv preprint arXiv:2605.17625*, 2026.
- [14] C.J. Hutto and E. Gilbert, "VADER: A Parsimonious Rule-Based Model for Sentiment Analysis," in *Proceedings of the International AAAI Conference on Web and Social Media*, 2014.
- [15] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pp. 3982-3992, 2019.
- [16] J. Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics*, pp. 4171-4186, 2019.
- [17] M. Ali, "Data Partitioning Strategies to Prevent Information Leakage in Deep Learning," *Springer Nature Applied Sciences*, vol. 7, pp. 15-29, 2025.
- [18] X. Chen et al., "When Text Embedding Meets Large Language Model: A Comprehensive Survey," *arXiv preprint arXiv:2502.12836*, 2025.
- [19] Mem0 Team, "Mem0: The Memory Layer for Personalized AI Agents," *GitHub*, 2024. [Online]. Available: <https://github.com/mem0ai/mem0>
- [20] G. Serra et al., "Federated continual learning goes online: Uncertainty-aware memory management," in *Proceedings of the International Conference on Learning Representations*, 2025.